

Penpicha Kongsup

System Analyst · Research & Data Background

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Analytical professional with 2+ years of experience working at the intersection of data, processes, and people in a complex, regulated environment. Skilled at understanding how systems and workflows operate end-to-end — identifying requirements, mapping processes, spotting gaps, and translating between technical and non-technical stakeholders. Combines a structured research mindset with hands-on technical skills in R, Python, and SQL. A self-directed learner who builds understanding from first principles.

EXPERIENCE

Research Assistant & Data Analyst

Feb 2024 – Present

Obesity Research Group · Siriraj Hospital, Mahidol University · Bangkok

- Analyzed end-to-end research workflows to identify process gaps, redundancies, and compliance risks — designed corrective procedures to improve system reliability and data integrity.
- Gathered and translated requirements from physician-investigators into structured study designs, statistical analysis plans, and operational protocols.
- Mapped and documented clinical data flows; maintained process consistency across cross-functional teams under strict GCP (ICH-E6 R3) and IRB compliance.
- Designed data collection systems and quality control frameworks; performed data validation, cleaning, and transformation using R and Python.
- Served as central communication point between clinical teams, ethics committees, and research leads — synthesizing information into clear, actionable documentation.
- Evaluated feasibility of proposed research designs against available data, resources, and constraints; provided structured recommendations balancing validity with operational reality.
- Managed IRB/EC documentation lifecycle: scoped requirements, drafted submissions, tracked status, and coordinated amendments with full audit trails.

Research Intern — Erasmus Mundus+

Sep 2021 – Jul 2022

University of Padova · Italy

- Understood existing workflows, adapted to new protocols, and contributed structured analysis within an international research system.
- Designed and standardised experimental protocols; identified procedural inconsistencies and proposed refinements to improve data quality.
- Managed multi-variable datasets from chemical composition and digestibility studies.

Researcher (Master's Degree Project)

2020 – 2021

Kasetsart University · Bangkok

- Designed in vitro and in vivo experimental systems from scratch — defining inputs, outputs, data collection logic, and success criteria.
- Managed full data lifecycle across concurrent projects: collection, cleaning, analysis, and interpretation.

SKILLS

Systems	Requirements gathering · Process mapping · Workflow design · Gap analysis · Data flow documentation · Feasibility assessment
Data & Tech	R · Python (Pandas, NumPy) · SQL · SQLite · MongoDB · Data Transformation · Data Visualization · Machine Learning
Tools	Google Sheets · GraphPad Prism · MS Office · HTML · CSS · Prompt Engineering
Domain	GCP ICH-E6(R3) · IRB/EC process · Regulatory compliance · Clinical data workflows

Soft Skills Analytical thinking · Stakeholder communication · Structured problem-solving · Self-directed learning

CERTIFICATIONS

GCP ICH-E6(R3) Training Course	Siriraj Hospital, Mahidol University	2026
Certified Professional Data Analyst	DataRockie	2026
Data Science Bootcamp 12	DataRockie	2026
16s rRNA Gene Sequencing & Analysis	Omics Sciences & Bioinformatics Center	2025
Gemini Certificate Educator	Google	2025
Basic Data Science	AI Association of Thailand	2025
AI Engineer	AI Association of Thailand	2025
Essential SQL for Everyone	BornToDev	2025
MongoDB for SQL Experts	MongoDB	2025
Monitor, IRB Member (CITI)	CITI Program	2024
Auditor, IRB Chair (CITI)	CITI Program	2024
Data Safety & Monitoring Boards	The Global Health Network	2024
Introduction to Research Ethics	The Global Health Network	2024

EDUCATION

M.Sc. Animal Health & Biomedical Sciences 2021 – 2022

Kasetsart University · Thesis-based · Erasmus Mundus+ Exchange, University of Padova, Italy

- Published 2 papers in international peer-reviewed journals (2022, 2025).
- Thesis: multi-variable experimental analysis — designing data collection systems, running statistical models, and interpreting results.

B.Sc. Biology 2017 – 2020

Kasetsart University, Faculty of Science

- Senior project: protein source screening using in vitro digestibility techniques — designed evaluation framework, analysed data, presented findings.
- Presented at KUVIC-AiSVS 2020 international conference.